

Unit IV: Electromagnetic Induction and Alternating Currents

CHAPTER-6: ELECTROMAGNETIC INDUCTION

GIST OF CHAPTER:

Electromagnetic induction; Faraday's laws, induced EMF and current; Lenz's Law, Self and mutual induction.

CONTENT/ CONCEPTS-

Magnetic flux - Total number of magnetic lines of forces passing normally through a given surface is called magnetic flux.

$$\phi = \vec{B} \cdot \vec{A} = BA \cos \theta$$

Electromagnetic induction - Phenomenon of production of induced emf (and hence induce current) due to change of magnetic flux linked with the coil or conductor is called electromagnetic induction.

Faraday's law of electromagnetic induction -

First Law - Whenever magnetic flux linked with the close circuit changes, emf (and hence current) is induced in it. This induced emf last as long as the change in magnetic flux continuous.

Second law - The magnitude of induced emf is directly proportional to the rate of change of magnetic flux linked with circuit.

$$\epsilon = - \frac{d\phi}{dt}$$

$$\text{If coil consists of } N \text{ turns then } \epsilon = -N \frac{d\phi}{dt} = -N \frac{(\phi_2 - \phi_1)}{t}$$

Lenz law - The direction of induced emf is such that it always opposes the change in magnetic flux which produces it.

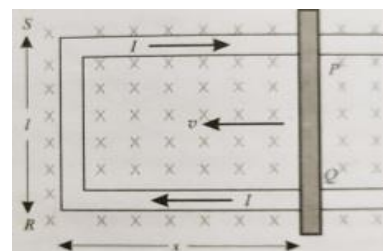
Motional emf - emf induced across the ends of a conductor due to its motion in magnetic field is called motional EMF.

EMF induced across the ends of conductor moving in a perpendicular magnetic field -

A] Magnetic flux enclosed by loop, $\Phi = B A = B l x$

According to Faraday's law, $\epsilon = - \frac{d\Phi}{dt} = - \frac{d}{dt} (B l x)$

$$\epsilon = + Blv \quad \text{where, } \frac{dx}{dt} = v$$



B] Induced current, $I = \frac{\epsilon}{R} = \frac{Blv}{R}$

$$\therefore \text{External force required, } F = B l I \sin 90^\circ = B \times \frac{Blv}{R} \times l = \frac{B^2 l^2 v}{R}$$

C] Power supplied by external force to maintain motion of movable arm,

$$P = Fv = \frac{B^2 l^2 v^2}{R}$$

D] Power dissipated as joule heating heat, $P = I^2 R = \frac{B^2 l^2 v^2}{R}$

Eddy current (Foucault current)- When a solid conducting material is placed in a changing magnetic field, induced current is set up inside the conductor. These current loops are just like eddies produced on water and hence known as eddy current.

Uses - 1] Magnetic braking in electric trains.

2] To produce heat in induction furnaces.

3] Electromagnetic damping

Disadvantages –

- 1] Heat up metallic core and dissipate energy.
- 2] Damage insulation of coil due to heat.
- 3] Always oppose motion.

Minimization of eddy current –

- 1] Using laminating iron core.
- 2] Using slotted iron blocks.

Self-induction - Self-induction is the property of coil by virtue of which it opposes any change in the current flowing through it by inducing emf in itself. (Also called inertia of electricity.)

Coefficient of self-induction(L) - At any instant, $\phi \propto I \Rightarrow \phi = L I$

$$\text{As } \epsilon = \frac{d\phi}{dt} \text{ (In mag.)} \Rightarrow \epsilon = L \frac{dI}{dt}$$

$$\text{If } \frac{dI}{dt} = 1 \text{ amp/sec then } \epsilon = L \Rightarrow$$

Definition - The self-inductance of coil is defined as emf induced in the coil due to unit rate of change of current in the same coil.

$$\text{Self-inductance of a long solenoid (L)} = \frac{\mu_0 (n l)^2 A}{l} = \mu_0 n^2 l A$$

If solenoid is filled with a material of relative permeability μ_r then $L = \mu_r \mu_0 n^2 l A$

$$\mu_m n^2 A l \quad \left(\text{As } \mu_r = \frac{\mu_m}{\mu_0} \right)$$

Factors on which self-inductance depends - 1] Number of turns

2] Area of cross section, 3] Permeability of core material

Mutual induction - Mutual induction is the phenomenon of production of induced emf in one coil due to change of current in the neighbouring coil.

Coefficient of mutual induction(M) –

At any instant,

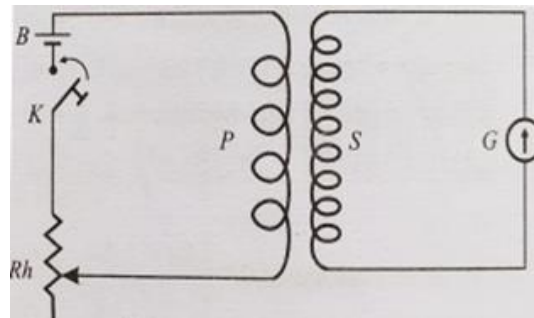
$$\phi \propto I_P$$

$$\Rightarrow \phi = M I_P$$

$$\text{As } \epsilon = \frac{d\phi}{dt} \text{ (In mag.) } \Rightarrow \epsilon = M \frac{dI_P}{dt}$$

$$\text{If } \frac{dI_P}{dt} = 1 \text{ amp/sec}$$

$$\text{then } \epsilon = M$$



Definition- Mutual inductance between primary and secondary coil is defined as the emf induced in secondary coil due to unit rate of change of current in primary coil.

Mutual inductance between two long solenoids of same length (M):-

$$M = \mu_0 n_1 n_2 A l$$

If solenoid is filled with a material of relative permeability μ_r then

$$M = \mu_r \mu_0 n_1 n_2 A l$$

$$= \mu_m n_1 n_2 A l$$

Factor on which mutual inductance depends –

- 1] Number of turns
- 2] Common cross-sectional area
- 3] Relative separation.
- 4] Relative orientation of two coil.
- 5] Permeability of core material

FORMULAE

Magnetic flux $\Phi = AB \cos\theta = \vec{A} \cdot \vec{B}$

Induced emf $\epsilon = -N \frac{d\Phi}{dt} = -N \frac{\Phi_2 - \Phi_1}{t}$

Induced across the ends of conductor moving in a perpendicular magnetic field

$$\epsilon = Blv$$

$$\text{Induced current, } I = \frac{\epsilon}{R} = \frac{Blv}{R}$$

$$\text{External force required, } F = \frac{B^2 l^2 v}{R}$$

$$\text{Power supplied by external force } P = \frac{B^2 l^2 v^2}{R}$$

$$\text{Power dissipated as joule heating heat, } P = I^2 R = \frac{B^2 l^2 v^2}{R}$$

Induce emf in a conducting rod rotated in a uniform magnetic field

$$\epsilon = \frac{1}{2} B R^2 \omega$$

Self-inductance of a long solenoid $L = \mu_0 n^2 l A$

If solenoid is filled with a material of relative permeability μ_r then

$$L = \mu_r \mu_0 n^2 l A = \mu_m n^2 A l$$

Mutual inductance between two long solenoids of same length $M = \mu_0 n_1 n_2 A l$

If solenoid is filled with a material of relative permeability μ_r then

$$M = \mu_r \mu_0 n_1 n_2 A l = \mu_m n_1 n_2 A l$$